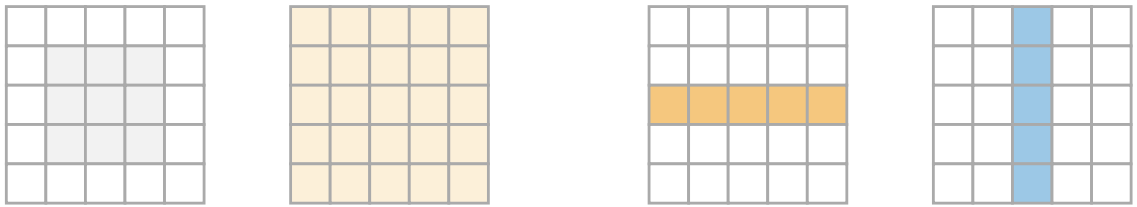
① Depthwise kernels (5×5 neighbourhood of one position)

ShapeAware **Directional**



3×3 5×5 1×5: horizontal 5×1: vertical

At P5 (stride 32) a 1×5 kernel covers about 160 input pixels: region-level shape, not pixel-level boundary.

② Computing the shape feature S

b 256 → DW 3×3 ‖ DW 5×5 → 1×1 → S_{shape} → S_h (1×5), S_v (5×1), S_{loc} (3×3) $S_{grad} = \sqrt{S_h^2 + S_v^2}$ → concat 4 → 1×1, concat S_{shape} → 1×1 → **S**

③ Gate: the shape branch modulates SS2D (one-way)

F_M (SS2D) → $F'_M = F_M \odot (1 + G)$ ← $G = \sigma(W_g * S + b_g)$, $G \in (0, 1)$

1 + G ∈ (1, 2): only amplifies, never switches a position off